

"We do use it, but not how hearing people think": How the Deaf and Hard of Hearing Community Uses Large Language Model Tools

Shuxu Huffman
Gallaudet University
Washington, District of Columbia
USA
shuxu.huffman@gallaudet.edu

Si Chen
University of Notre Dame
Notre Dame, Indiana, USA
schen34@nd.edu

Kelly Avery Mack
Department of Human Centered
Design and Engineering
University of Washington
Seattle, Washington, USA
kmack3@uw.edu

Haotian Su
Department of Automotive
Engineering
Clemson University
Greenville, South Carolina, USA
haotias@g.clemson.edu

Qi Wang
Business
Gallaudet University
Washington, District of Columbia
USA
qi.wang@gallaudet.edu

Raja Kushalnagar
Gallaudet University
Washington, District of Columbia
USA
raja.kushalnagar@gallaudet.edu

Abstract

Generative AI tools, particularly those utilizing large language models (LLMs), are increasingly used in everyday contexts. While these tools enhance productivity and accessibility, little is known about how Deaf and Hard of Hearing (DHH) individuals engage with them or the challenges they face when using them. This paper presents a mixed-method study exploring how the DHH community uses Text AI tools like ChatGPT to reduce communication barriers and enhance information access. We surveyed 80 DHH participants and conducted interviews with 9 participants. Our findings reveal important benefits, such as eased communication and bridging Deaf and hearing cultures, alongside challenges like lack of American Sign Language (ASL) support and Deaf cultural understanding. We highlight unique usage patterns, propose inclusive design recommendations, and outline future research directions to improve Text AI accessibility for the DHH community.

CCS Concepts

• **Human-centered computing** → **Empirical studies in accessibility**.

Keywords

Accessibility, LLM, ChatGPT, Deaf and Hard of hearing

ACM Reference Format:

Shuxu Huffman, Si Chen, Kelly Avery Mack, Haotian Su, Qi Wang, and Raja Kushalnagar. 2025. "We do use it, but not how hearing people think": How the Deaf and Hard of Hearing Community Uses Large Language Model Tools. In *Extended Abstracts of the CHI Conference on Human Factors in*

Computing Systems (CHI EA '25), April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3706599.3719785>

1 Introduction

Generative AI tools, such as OpenAI's ChatGPT and Microsoft Copilot, are widely used across professional and personal contexts, including email writing, programming, and accessibility support [22, 24, 41, 50]. Text-based generative AI, which we refer to as "Text AI"¹, powered by LLMs, is particularly popular for its ability to process natural language inputs and generate human-like text [22].

Text AI holds significant potential [22, 55] for the Deaf and Hard of Hearing (DHH) community, which is characterized by strong cultural identities and diverse linguistic backgrounds [2, 11, 57]. Many DHH individuals are bilingual, with sign language (e.g., ASL) as their primary language and written language (e.g., English) as their second language [57]. Text AI can help bridge language gaps by improving fluency and clarity in written English, reducing communication barriers in a predominantly hearing society. However, the lack of research on how DHH individuals use Text AI limits understanding of its benefits and challenges. This work addresses these gaps by exploring how the DHH community interacts with Text AI and identifying barriers to its use. Specifically, we investigate:

RQ1 How does the DHH community interact with and utilize Text AI tools?

RQ2 What are the challenges and opportunities of using Text AI in the DHH community?

To answer these questions, we conducted a survey with 80 DHH participants and interviewed 9 DHH participants using ASL and English study materials. Our analysis reveals that DHH participants exhibit mixed feelings and opinions about their use of Text AI. On the one hand, they acknowledged meaningful benefits that AI tools offer in overcoming communication barriers in a hearing-majority

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719785>

¹We use this term because there is no standard ASL sign for "LLM" or "Generative AI". However, in the Deaf community people refer to such tools as MESSAGE-AI or TYPE-AI so we chose a more general term that is closest to the interpretation.

society, including bridging Deaf and hearing cultures and increasing confidence in written English communication. On the other hand, they raised concerns about the tools' lack of DHH-friendly interaction techniques, education, guidance, and understanding of Deaf cultural nuances. Participants also offered valuable suggestions for improving Text AI tools to better meet the needs of DHH users.

2 Background and Related Work

2.1 Background of The DHH Community

The Deaf² community encompasses rich cultural and linguistic diversity, characterized by unique identities and experiences [2, 57, 64]. Scholars advocate for a postcolonial framework over the traditional medical model to emphasize positive aspects of Deaf culture, including concepts like Deafhood, DeafSpace, and Deaf gain [3, 21, 40]. However, not all DHH individuals identify with Deaf culture, as connections to Deaf identity often vary due to linguistic and cultural factors [11]. While this paper focuses on the broader DHH community, attention is also given to the Deaf community's cultural and linguistic context.

Sign language, particularly ASL, is a cornerstone of Deaf culture and identity, recognized as a fully developed language with its own grammar and vocabulary [66, 70]. Features like ASL gloss—a notation system used to represent individual signs in written form [71]—have proven effective in improving English literacy among Deaf children [67]. Additionally, DHH individuals adopt diverse strategies to communicate with hearing people, blending oral and visual methods [19]. Scholars highlight the need for culturally responsive research methodologies to meet the educational and cultural needs of DHH individuals [11, 63].

2.2 Unique Challenges The DHH Community Faces

The DHH community faces issues such as societal bias, discrimination, and ableism, many of which are experienced in other disability communities [14, 45, 49]. However, there are also challenges that are specific to the DHH community.

2.2.1 Difficulties in Language Acquisition. Deaf children face significant challenges in acquiring spoken languages due to limited auditory input and reliance on visual communication, compounded by educational resources designed for hearing learners [15, 39, 68]. The historical emphasis on oralism over sign language use has resulted in language deprivation during critical developmental periods, leading to long-term academic, social, and emotional challenges [25, 29, 30, 33]. Marginalizing ASL while prioritizing English without accessible instruction exacerbates difficulties in language acquisition, particularly with English fluency and grammar [10, 58]. Moreover, unlike spoken languages, sign language exists in a visual format without a written form, further complicating language learning. Researchers emphasize addressing these barriers through culturally informed educational approaches [12, 39, 59].

2.2.2 Discrimination and Trauma. Discrimination against DHH individuals often stems from linguistic bias, misconceptions about sign language, and communication barriers, resulting in marginalization across workplaces, education, and social settings [23, 37]. Daily communication barriers can lead to frustration, trauma, and exclusion [16, 42]. Audism, the belief in the superiority of hearing over deafness, further marginalizes DHH individuals, particularly sign language users, by reinforcing systemic inequalities in employment and social interactions [54].

The DHH community is also uniquely vulnerable to Information Deprivation Trauma (IDT), characterized by emotional distress caused by a lack of access to critical information [62]. This is especially pronounced in a hearing-dominated society where spoken-language information is often inaccessible without alternatives such as sign language or captions. IDT disproportionately affects Deaf individuals, hindering their ability to understand, prepare for, or respond to significant situations [61, 62].

Text AI, as a language tool [22, 31, 46, 55], has the potential to mitigate these challenges. While research has shown that AI-assisted learning can significantly enhance communication abilities and reduce anxiety among non-native English speakers [31, 46], little is known about its impact on individuals who experience both language minority status and deafness. This study explores the potential of Text AI to address these challenges.

2.2.3 The DHH Community in the AI Era. The rise of AI offers opportunities and challenges for the DHH community. While tools to improve English grammar exist [9], biases in AI—rooted in non-representative datasets and assumptions by hearing researchers—risk systematically excluding Deaf users [18, 28]. Advances in tools such as automatic captions and sign language recognition and translation have improved accessibility in work and education [1, 8, 13, 36, 56]. However, little research explores how DHH individuals use generative AI tools. This study examines their usage patterns, benefits, challenges, and barriers with these tools.

3 Methods

Our study combines surveys for broad insights and interviews for deeper perspectives. Surveys identify patterns, while interviews explore usage in depth. This approach triangulates findings, capturing both trends and nuanced experiences.

The research team included Deaf and hearing members with varying signing abilities.

3.1 Survey

We conducted an online survey targeting DHH individuals. Participants met the following criteria: 18 years or older, U.S.-based, self-identified as DHH, and fluent in ASL, English, or both. Recruitment was conducted through social media, email lists, a Deaf non-profit organization, a university, and snowball sampling [60]. To ensure accessibility, materials, including the consent form, were provided in both ASL and English [44]. Multiple-choice questions used straight-forward language, while open-ended questions allowed responses via text or uploaded ASL videos. At the survey's start, participants were given a definition of Text AI, explaining it as a program that generates text in response to user prompts, with examples like ChatGPT, Copilot, and Gemini. Participants

²Capital "D" Deaf refers to individuals who identify with Deaf culture and use sign language, while deaf refers to the physical condition of hearing loss without necessarily identifying with the Deaf community [2, 57].

were entered into a raffle to win one \$20 gift card for every 10 participants.

We received 90 responses, excluded 10 bot responses, and tabulated the remaining 80 valid responses. One participant opted for a video chat, and a Deaf researcher conducted the session. The average participant age was 37.5 years ($SD = 12.3$). Among respondents, 55 identified as Deaf, 13 as Hard of Hearing, 8 as deaf, 1 as Deaf-Blind, 1 as Deaf and Dumb, and 2 as other, specifying deaf/HoH³ and Deaf/HoH. ASL was the primary communication language for 52 participants, 22 used English, and 3 used both depending on the situation. Additional languages reported included Armenian, Farsi, and Chinese.

The survey questions (provided in the Appendix) were inspired by our RQs and relevant literature [32, 43]. The survey comprised six sections, structured around five key variables we aimed to analyze: 1. frequency of Text AI use, 2. types of tasks for which Text AI is used, 3. reduction in anxiety or perceived discrimination, 4. satisfaction with Text AI assistance, and 5. the difficulties and challenges encountered while using Text AI.

3.2 Semi-structured Interview

We conducted semi-structured interviews ($N = 11$) to explore how DHH individuals use Text AI tools in daily life. Since two participants also completed the survey, we excluded their data from the analysis, resulting in a final sample of $N = 9$. Among the participants, five self-identified as Deaf and four as Hard of Hearing. Participants, aged 20 to 42, represented diverse genders, ethnicities (Eritrean, Caucasian, Hispanic, Filipino, and Middle Eastern), and communication preferences, including ASL, other sign languages, English, or a combination, reflecting varied experiences within the DHH community. Recruitment was conducted through a university email list and expanded via snowball sampling [60] with the same recruitment criteria as the survey.

Interviews included task-based interactions with ChatGPT, the most commonly used Text AI tool reflected in our survey. Participants completed three tasks: (1) drafting an email to a professor requesting a homework extension, (2) addressing a group teammate’s tardiness in an email, and (3) inquiring about an internship opportunity with a university recruiter. Since most participants were recruited from a university, these tasks were tailored to reflect common scenarios in university life. Follow-up questions assessed their satisfaction with ChatGPT’s responses, typical usage patterns, and additional challenges or questions. Interviews were conducted over Zoom, lasting approximately 30 minutes, with shorter durations for some participants. To ensure accessibility [44], ASL interpreters and Zoom’s auto-captioning were provided as needed. Participants were encouraged to ask questions and seek clarification at any time and received gift cards as compensation with an hourly rate of \$25.

3.3 Interview and Survey Analysis

A Deaf researcher translated the ASL interview responses into English. Two researchers conducted thematic analysis [5, 6] by independently developing codes, reconciling differences, and mutually

verifying and discussing the final codes. Together, they collaboratively identified key themes to represent the findings. For the survey data, quantitative analysis examined participants’ feelings with and without using Text AI and explored how their primary language influenced perceptions. While comparisons such as Deaf versus Hard of Hearing were possible, the analysis focused on primary language (ASL or English) due to the data distribution and the nuanced spectrum of identities within the community, as reflected in responses to open-ended identity questions.

4 Results and Discussion

We first present usage patterns and experiences with Text AI systems, focusing on their role in enhancing communication confidence and bridging cultural gaps (RQ1). We also address concerns such as loss of cultural nuances and suggestions for improvement (RQ2). Our analysis integrates insights from open-ended survey responses, interviews, and statistical data, with interview participants labeled as “I#” and survey respondents as “S#.”

4.1 RQ1-How Does The DHH Community Interact With and Utilize Text AI Tools?

Beyond common Text AI tasks within the general population (writing emails, looking up facts), we identified several themes regarding how DHH individuals specifically interact with and use Text AI tools, presenting qualitative results first, followed by quantitative findings. While some usage patterns align with those of non-native English speakers, others are uniquely shaped by the intersections of language barriers, access needs, and the visual nature of sign language.

4.1.1 Text AI for Language Support: Uneven Gains in Confidence and Learning. Text AI tools help reduce language barriers for the DHH community, especially when communicating with English-speaking hearing individuals. Participants noted that using non-standard English, like ASL gloss, often leads to misunderstandings, criticism, and discrimination, affecting confidence [23]. For example, S2 shared, “Deaf people write in ‘broken’ English... hearing people struggle to understand, but ASL users pick it up easily because it’s written in ASL gloss.” S79 also shared their experience in school: “I always forgot to add the ‘-s’, ‘-ing’, commas... teachers were always mad at me, and I lost my confidence. I never finished college English.”

Participants reported that Text AI boosted their confidence in writing by providing nonjudgmental corrections and improving grammar, vocabulary, tone, and fluency. As S78 expressed, “It makes using English easier and better than before... it boosts my confidence and makes me feel more like myself.” Another participant (S79) valued Text AI over a tutor in language learning, saying, “I’m not embarrassed. I just get my English corrected.”

People who prefer ASL differ from most other non-native English users, because ASL is a visual language with no direct written translation. Many non-native English speakers can query Text AI for English support in their preferred language⁴. Therefore, one might expect their use of Text AI to differ from that of non-signing, non-native English speakers. Interestingly, however, their usage patterns showed significant alignment [31, 46]. Still, as we describe

³HoH refers to Hard of Hearing.

⁴ChatGPT, at the time of this study, supports over 50 languages [53].

in the next section, the consequences of using non-fluent English are magnified for DHH individuals as they navigate ableism and advocate for accommodations in their daily lives. Furthermore, usage patterns for DHH individuals may look more distinct if Text AI allowed for signing inputs and outputs, which we discuss more in section 4.2.

Finally, while most users experienced benefits from using Text AI, a few reported decreased comfort. Figure 1(a) shows that some participants, particularly those already comfortable with written English, experienced this decline. While some participants appreciated that Text AI made their English more aligned with their intentions, others found the responses unnatural. This contradiction may stem from the DHH community's diverse cultural and linguistic backgrounds. Participants who described AI responses as "fake" (I14) or "unnatural" (I13) were predominantly college-educated, which suggests that higher English proficiency might increase sensitivity to linguistic nuances. Participants with high English proficiency raised a concern for the broader DHH community: Text AI may lead DHH users with lower English proficiency to automatically trust its outputs, leading to misinformation, a trend also observed in Automatic Speech Recognition (ASR) technologies [4].

Survey data supports these findings. Among the 68 users with experiences of using Text AI, 55.9% reported Text AI improved fluency, 50.0% organization, 48.5% vocabulary, 42.6% grammar, 42.6% tone and voice, and 45.6% overall confidence. Satisfaction levels varied, with 58.8% feeling satisfied or very satisfied with its assistance in improving English proficiency. Although the Wilcoxon Signed-Rank Test revealed no statistically significant difference in comfort levels before and after using Text AI ($p = 0.08$), we observed trends suggesting potential improvements. Without Text AI, 63.2% of users felt comfortable or very comfortable using written English to communicate, compared to 82.4% with Text AI, while those feeling somewhat uncomfortable or very uncomfortable dropped from 14.7% to 8.8%. Likert-scale data indicates that ASL users' comfort levels increased (3.6 to 4.0), while English users' levels slightly declined (4.6 to 4.5).

4.1.2 Bridging Cultural Gaps and Easing Communication With Hearing People. The Deaf community faces unique cultural challenges beyond language barriers, including communication needs and navigating differences with hearing culture. Our study found that Text AI can serve as a cultural mediator, bridging gaps between Deaf and hearing cultures in contexts ranging from everyday interactions to addressing ableism, and alleviating communication stress when interacting with hearing individuals.

One participant (S76) shared how ChatGPT helped bridge communication gaps with their hearing family, who often misunderstood their Deaf perspective: *"My Deaf perspective and opinions just don't resonate with my family."* Living between Deaf and hearing cultures, S76 found that carefully wording their messages reduced conflict but described this effort as *"exhausting as heck."* They spent 15 years away from their family due to communication challenges. After returning, they began using ChatGPT to reword their messages, requesting it to be *"casual but sound firm."* They explained, *"Especially when my emotions are high, I type out what I need to express to ChatGPT (unfiltered), and it filters it in a way that is more*

acceptable to express." This tool became a vital intermediary, with the participant noting, *"Somehow, every single time, ChatGPT wins. I use what it suggests, and things go more smoothly... And I feel closer to my family. There is less conflict."*

Participants highlighted cultural differences in communication styles as a challenge, particularly the perceived bluntness of Deaf culture compared to the more indirect norms of hearing culture [26]. Text AI tools were valued by our participants (I7, S59, S75, S22) for helping to adjust tone and reduce misunderstandings in these interactions. These tools also supported participants in navigating sensitive conversations about access and ableism. One participant (S44) explained, *"I use ChatGPT to make my emails come off as more 'polite' when I need to address access issues or ableism with my coworkers and supervisors."* Text AI tools helped reduce the emotional burden of advocating for accommodations and addressing discrimination, which is often exhausting for DHH individuals.

Our quantitative data also indicates that Text AI significantly reduced communication anxiety experienced by DHH individuals when interacting with hearing people (Fig. 1(b)). The proportion of users experiencing daily or weekly anxiety decreased from 14.7% without Text AI to 5.9% with its assistance, while those reporting never or rarely feeling anxious increased from 63.2% to 83.8%. The Stuart Maxwell test confirmed that the anxiety level reduction was statistically significant ($p < .01$). This finding demonstrates Text AI's effectiveness to ease cross-cultural communication.

4.1.3 Fact-Checking and Simplifying Information Retrieval. While Text AI tools face criticism for errors and aberrations [22], participants highlighted their unique benefits for DHH users, particularly in clarifying ambiguous queries and reducing cognitive load during the process of gathering information. Unlike Google, Text AI's conversational format allows users to refine their language, making information retrieval more accessible for some DHH individuals. For instance, S79 described using one Text AI tool to clarify their query in English and another to seek information: *"I get my thoughts together, use ChatGPT to make my English, then give to Gemini and ask 'true or false?'"*

Summarization was another key feature for participants, enabling them to condense dense texts and meeting transcripts. As S8 shared, *"I give a complicated wall of text and ask it to simplify for me,"* reducing the mental fatigue common among DHH individuals when processing complex content. Participants also used Text AI to summarize meeting transcripts, mitigating issues like rapid speech of multiple parties and captioning errors [47, 48]. Text AI may further alleviate Information Deprivation Trauma (IDT) by providing accessible tools for forming questions, verifying facts, simplifying information, and bridging gaps in critical knowledge. However, its effectiveness relies on accuracy and cultural sensitivity, emphasizing the need for further refinements.

4.2 RQ2-What Are the Challenges and Opportunities of Using Text AI in the DHH Community?

We identify the challenges the DHH community faces when using Text AI, followed by suggestions and opportunities to address these challenges and improve the tools' accessibility and inclusivity.

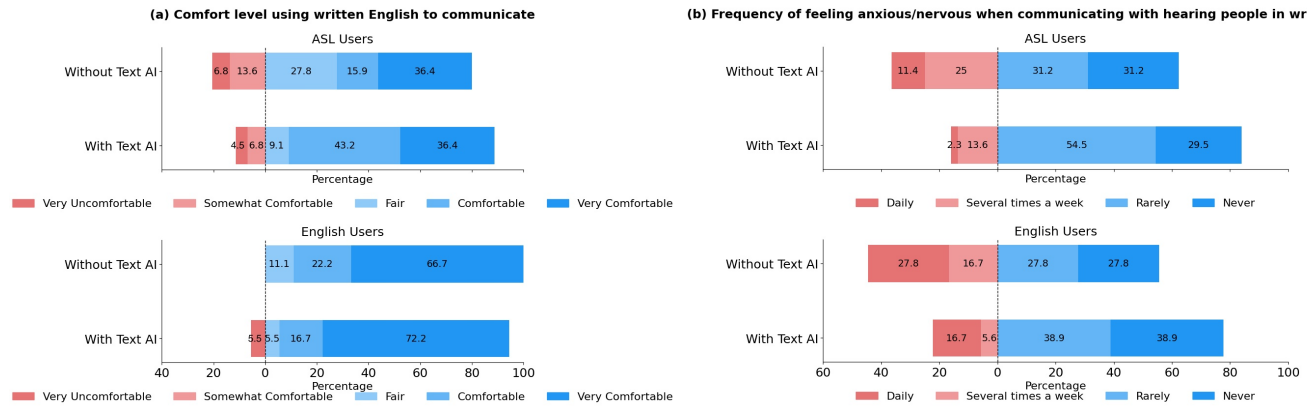


Figure 1: The diagrams show data from Text AI users with ASL or English as their primary language (N = 62), comparing comfort levels and anxiety frequencies in written communication with and without Text AI. Data from 12 non-users, 3 participants with other primary languages, and 3 participants who identified both ASL and English as their primary languages are excluded. Diagram (a) shows changes in comfort levels using written English for ASL users (N = 44) and English users (N = 18). Diagram (b) presents changes in anxiety frequency when communicating with hearing individuals in writing. Text AI increased comfort levels for ASL users but slightly decreased them for English users in (a), while reducing anxiety frequencies for both groups in (b).

4.2.1 Challenge 1: Crafting Effective Prompts in English. Among participants that have previously used Text AI, 44.1% reported difficulties in crafting effective prompts, 23.5% experienced technical issues (such as network connectivity issues), 22.1% noted limited sign language support, 14.7% expressed doubts about AI’s utility, and 10.3% found responses difficult to understand. The primary challenge was shaping questions in English, akin to prompt engineering, to achieve specific or accurate results. As one participant (S53) noted, “As a native ASL user, it takes time to generate the right prompt because it’s text-only.” This highlights the effort required for ASL users to translate goals into effective prompts, especially when conveying nuanced or complex text input.

Suggestion: Text AI Should Support ASL and Diverse Language Needs Most Text AI tools rely on moderate-to-high English proficiency of users, limiting accessibility for users with lower proficiency. Participants recommended features like ASL video explanations using virtual avatars to enhance accessibility and provide tailored feedback for prompt improvements. While ASL recognition and generation through avatars remain imperfect [17, 38, 52, 65], a practical first step could involve training Text AI on ASL gloss. More generally, participants highlighted the need for Text AI to better understand and improve prompts written in ASL gloss. As S22 suggested, “Maybe another option is to ask the LLM to respond in ASL grammar order.” Training Text AI to recognize ASL gloss would require specialized datasets sourced from the DHH community and guided by fairness and disability-first principles [69]. This adaptation could help the tool “better learn to understand prompts from Deaf users” (S2), making it more inclusive and effective, and serve as a valuable language-learning aid, reducing the cognitive burden of translating between ASL and English.

Suggestion: DHH-Focused Tutorials and Instructional Design for Text AI Participants stressed the importance of DHH-friendly instructions to address language deprivation and teach effective use of Text AI. Suggestions included offering classes tailored to DHH users in schools and colleges (S64) and creating resources explaining “how Text AI can help particularly Deaf/HoH users” (S50). Developing such tutorials requires significant resources and expertise in both Text AI and the DHH community. While institutions like higher education have created tutorials for instructors and students [35], similar efforts for the DHH community are lacking and should be prioritized. When top-down approaches fall short, bottom-up methods, like crowdsourcing, may empower underrepresented groups to lead efforts and share knowledge [34] and should be studied in future research.

4.2.2 Challenge 2: Perceived Shallow Understanding of The DHH community by Text AI. Participants felt that Text AI tools lack a nuanced understanding of the Deaf community’s diversity, where deafness may be viewed as anything from a medical condition to a cultural identity. This gap limits the AI’s ability to address important historical and social contexts. For example, S79 compared ChatGPT and Copilot’s explanations of the word “dumb” and noted that while ChatGPT provided historical context about its relevance to disability, Copilot omitted this information⁵. Although the term “deaf and dumb” is now considered outdated and inappropriate, it remains part of Deaf history, and its historical context should still be shared as a fact when such terms arise. Participants also criticized Text AI’s adherence to standardized notions of “professionalism,” which they perceived as rooted in “white hearing culture.” For example, S44 remarked that the overly formal phrasing, such as “I hope this email finds you well,” felt inauthentic and disconnected from their natural communication style. This disconnect

⁵A comparison of the chat histories is included in the appendix.

highlights a need for AI to account for diverse cultural norms and communication preferences.

Suggestion: Incorporating Deaf Community Input into AI Models Beyond ASL support, participants emphasized the need for Text AI to integrate Deaf cultural knowledge and linguistic nuances through training on data from Deaf individuals. As S44 noted, “It needs further representation of culturally Deaf ideas, practices, and communication in the data sets.” Incorporating cultural awareness would help Text AI interpret terms like “dumb” accurately and foster more inclusive interactions. Further research is needed to explore how to incorporate Deaf community input and address discrepancies between LLM understanding and real-world experiences. Additionally, integrating existing research, such as how DHH reading comprehension varies with phonological and language awareness during key developmental phases [20], is essential for improving personalized interactions for individual users.

Suggestion: More Research on How and Why DHH Individuals Disclose to Text AI A few participants shared cultural, linguistic, and accessibility details with Text AI, finding it helpful for refining prompts. For example, S76 explained, “In ChatGPT it has a memory, I have it remembered I’m an expert user of English and ASL.” None of our participants intentionally avoided discussing their English proficiency or background when using Text AI. Our findings may be skewed towards individuals who see deafness as a cultural identity rather than a medical condition, with 68.8% self-identifying as Deaf. Moreover, research shows that identity evolves over time; for instance, DHH adolescents often integrate other significant aspects, such as ethnicity and culture, into their context-dependent d/Deaf identity [7]. Future research should explore how DHH individuals self-identify with AI and whether they adapt identity cues to receive tailored responses.

4.3 Limitations

Despite ASL translation and video submission options, the survey was not fully accessible to all participants. One participant requested a video call, and video uploads posed challenges for some [43]. Accessibility for DeafBlind users was limited, with only one participant despite multiple invitees; ProTactile⁶ options could have improved inclusion. The survey was primarily distributed within the Deaf community, potentially underrepresenting Hard of Hearing individuals, and university-based recruitment skewed the sample toward college students. Future research should broaden recruitment to include a more diverse DHH population and those less affiliated with Deaf culture.

5 Conclusion

This study explored DHH individuals’ experiences with Text AI tools, highlighting both benefits and barriers. Participants used Text AI to improve communication with hearing individuals through English support and cultural bridging, alleviating much of the effort DHH people typically invest to facilitate communication. While Text AI helps reduce gaps between Deaf and hearing communities, challenges persist, including difficulties with English prompts, limited understanding of Deaf culture, and the need for Deaf-specific instructional materials. Participants called for ASL input/output

support, better adaptation to ASL grammar, and AI trained on Deaf culture. These findings provide insights into the unique ways the DHH community interacts with Text AI and offer recommendations to improve these tools for accessibility and inclusion. Future research should focus on developing AI that better serves underserved and underrepresented communities.

Acknowledgments

We thank Dr. Emma J. McDonnell, Dr. Han Zhang, and Dr. Christopher Hayes for their valuable feedback, as well as the DHH community for their input and support. This study was supported by the National Science Foundation (NSF) Awards #2150429 and #2131524. This study was approved by the Gallaudet University Institutional Review Board; IRB-FY24-122.

References

- [1] Mariana Arroyo Chavez, Bernard Thompson, Molly Feanny, Kafayat Alabi, Minchan Kim, Lu Ming, Abraham Glasser, Raja Kushalnagar, and Christian Vogler. 2024. Customization of closed captions via large language models. In *International Conference on Computers Helping People with Special Needs*. Springer, 50–58.
- [2] Yael Bat-Chava. 2000. Diversity of deaf identities. *American annals of the deaf* (2000), 420–428.
- [3] H-Dirksen L Bauman and Joseph J Murray. 2014. *Deaf gain: Raising the stakes for human diversity*. U of Minnesota Press.
- [4] Larwan Berke, Sushant Kafle, and Matt Huenerfauth. 2018. Methods for evaluation of imperfect captioning tools by deaf or hard-of-hearing users at different reading literacy levels. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–12.
- [5] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.
- [6] Virginia Braun and Victoria Clarke. 2019. Reflecting on reflexive thematic analysis. *Qualitative research in sport, exercise and health* 11, 4 (2019), 589–597.
- [7] Timothy J Byatt, Jill Duncan, Kerry Dally, and Louise de Beuzeville. 2021. Evolving identities of adolescents who are deaf or hard of hearing. *American Annals of the Deaf* 166, 1 (2021), 5–30.
- [8] Necati Cihan Camgoz, Oscar Koller, Simon Hadfield, and Richard Bowden. 2020. Sign language transformers: Joint end-to-end sign language recognition and translation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10023–10033.
- [9] Joanna E Cannon, Susan R Easterbrooks, Phill Gagné, and Jennifer Beal-Alvarez. 2011. Improving DHH students’ grammar through an individualized software program. *Journal of Deaf Studies and Deaf Education* 16, 4 (2011), 437–457.
- [10] Joanna E Cannon and Susannah Kirby. 2013. Grammar structures and deaf and hard of hearing students: A review of past performance and a report of new findings. *American annals of the deaf* 158, 3 (2013), 292–310.
- [11] Joanna E Cannon and John L Luckner. 2016. Increasing cultural and linguistic diversity in deaf education teacher preparation programs. *American Annals of the Deaf* 161, 1 (2016), 89–103.
- [12] Si Chen, Haocong Cheng, Suzy Su, Stephanie Patterson, Raja Kushalnagar, Qi Wang, and Yun Huang. 2024. Customizing Generated Signs and Voices of AI Avatars: Deaf-Centric Mixed-Reality Design for Deaf-Hearing Communication. *arXiv preprint arXiv:2410.01604* (2024).
- [13] Si Cheng, Shuxu Huffman, Qingxiaoyang Zhu, Haotian Su, Raja Kushalnagar, and Qi Wang. 2024. “Real Learner Data Matters” Exploring the Design of LLM-Powered Question Generation for Deaf and Hard of Hearing Learners. *arXiv preprint arXiv:2410.00194* (2024).
- [14] Adrienne Colella and Dianna L Stone. 2013. Workplace discrimination toward persons with disabilities: A call for some new research directions. In *Discrimination at work*. Psychology Press, 227–253.
- [15] Kata Csizér and Edit H Kontra. 2020. Foreign language learning characteristics of deaf and severely hard-of-hearing students. *The Modern Language Journal* 104, 1 (2020), 233–249.
- [16] Maitraye Das, Abigale Stangl, and Leah Findlater. 2024. “That comes with a huge career cost”: Understanding Collaborative Ideation Experiences of Disabled Professionals. *Proceedings of the ACM on Human-Computer Interaction* 8, CSCW1 (2024), 1–28.
- [17] Aashaka Desai, Lauren Berger, Fyodor Minakov, Nessa Milano, Chinmay Singh, Kriston Pumphrey, Richard Ladner, Hal Daumé III, Alex X Lu, Naomi Caselli, et al. 2024. ASL citizen: a community-sourced dataset for advancing isolated sign language recognition. *Advances in Neural Information Processing Systems* 36 (2024).

⁶ProTactile is a tactile language used by the DeafBlind community [27, 51].

- [18] Aashaka Desai, Maartje De Meulder, Julie A Hochgesang, Annemarie Kocab, and Alex X Lu. 2024. Systemic Biases in Sign Language AI Research: A Deaf-Led Call to Reevaluate Research Agendas. *arXiv preprint arXiv:2403.02563* (2024).
- [19] Ewa Domagala-Zyśk and Anna Podlowska. 2019. Strategies of oral communication of deaf and hard-of-hearing (D/HH) non-native English users. *European Journal of Special Needs Education* 34, 2 (2019), 156–171.
- [20] Susan R Easterbrooks, Amy R Lederberg, Shirin Antia, Brenda Schick, Poorna Kushalnagar, Mi-Young Webb, Lee Branum-Martin, and Carol McDonald Connor. 2015. Reading among diverse DHH learners: What, how, and for whom? *American Annals of the Deaf* 159, 5 (2015), 419–432.
- [21] Claire Edwards and Gill Harold. 2014. DeafSpace and the principles of universal design. *Disability and Rehabilitation* 36, 16 (2014), 1350–1359.
- [22] Stefan Feuerriegel, Jochen Hartmann, Christian Janiesch, and Patrick Zschech. 2024. Generative ai. *Business & Information Systems Engineering* 66, 1 (2024), 111–126.
- [23] Sally T Freeman. 1989. Cultural and linguistic bias in mental health evaluations of deaf people. *Rehabilitation Psychology* 34, 1 (1989), 51.
- [24] Jon E Froehlich, Chu Li, Maryam Hosseini, Fabio Miranda, Andres Sevtuk, and Yochai Eisenberg. 2024. The Future of Urban Accessibility: The Role of AI. (2024).
- [25] Neil S Glickman and Wyatt C Hall. 2018. *Language deprivation and deaf mental health*. Routledge.
- [26] Blaine Goss. 2003. Hearing from the deaf culture. *Intercultural communication studies* 12, 2 (2003), 1–17.
- [27] AJ Granda and Jelica Nuccio. 2018. ProTactile Principles. <http://wasli.org/wp-content/uploads/2018/05/PT-Principles-Movies-Final.pdf>
- [28] Jan Gugenheimer, Katrin Plaumann, Florian Schaub, Patrizia Di Campli San Vito, Saskia Duck, Melanie Rabus, and Enrico Rukzio. 2017. The impact of assistive technology on communication quality between deaf and hearing individuals. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing*. 669–682.
- [29] Matthew L Hall, Wyatt C Hall, and Naomi K Caselli. 2019. Deaf children need language, not (just) speech. *First Language* 39, 4 (2019), 367–395.
- [30] Wyatt C Hall. 2017. What you don’t know can hurt you: The risk of language deprivation by impairing sign language development in deaf children. *Maternal and child health journal* 21, 5 (2017), 961–965.
- [31] Kotaro Hayashi and Takeshi Sato. 2024. The effectiveness of ChatGPT in enhancing English language proficiency and reducing second language anxiety (L2). In *WorldCALL2023: Official Conference Proceedings*. 201–208.
- [32] Tracie Heskett. 2001. *Using the Six Trait Writing Model*. Teacher Created Resources.
- [33] Iain Hutchison. 2007. Oralism: A sign of the times? The contest for deaf communication in education provision in late nineteenth-century Scotland. *European Review of History—Revue européenne d’Histoire* 14, 4 (2007), 481–501.
- [34] Laura Inzerillo and Cettina Santagati. 2016. Crowdsourcing cultural heritage: from 3D modeling to the engagement of young generations. In *Digital Heritage. Progress in Cultural Heritage: Documentation, Preservation, and Protection: 6th International Conference, EuroMed 2016, Nicosia, Cyprus, October 31–November 5, 2016, Proceedings, Part I* 6. Springer, 869–879.
- [35] Yueqiao Jin, Lixiang Yan, Vanessa Echeverria, Dragan Gašević, and Roberto Martinez-Maldonado. 2024. Generative AI in Higher Education: A Global Perspective of Institutional Adoption Policies and Guidelines. *arXiv preprint arXiv:2405.11800* (2024).
- [36] Hamid Reza Vaezi Joze and Oscar Koller. 2018. Ms-asl: A large-scale data set and benchmark for understanding american sign language. *arXiv preprint arXiv:1812.01053* (2018).
- [37] Peter Jüni, Franziska Holenstein, Jonathan Sterne, Christopher Bartlett, and Matthias Egger. 2002. Direction and impact of language bias in meta-analyses of controlled trials: empirical study. *International journal of epidemiology* 31, 1 (2002), 115–123.
- [38] Lee Kezar, Jesse Thomason, Naomi Caselli, Zed Sehyr, and Elana Pontecorvo. 2023. The Sem-Lex Benchmark: Modeling ASL Signs and Their Phonemes. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility*. 1–10.
- [39] Edit H Kontra. 2017. The foreign-language learning situation of Deaf adults: An overview. *Journal of Adult Learning, Knowledge and Innovation* 1, 1 (2017), 35–42.
- [40] Paddy Ladd. 2005. Deafhood: A concept stressing possibilities, not deficits. *Scandinavian Journal of Public Health* 33, 66_suppl (2005), 12–17.
- [41] Tim Laquintano, Carly Schnitzler, and Annette Vee. 2023. An introduction to teaching with text generation technologies. *TextGenEd: Teaching with text generation technologies* (2023).
- [42] Yidan Ma, Weifeng Xue, Qin Liu, and Yin Xu. 2022. Discrimination and deaf adolescents’ subjective well-being: The role of deaf identity. *The Journal of Deaf Studies and Deaf Education* 27, 4 (2022), 399–407.
- [43] Kelly Mack, Danielle Bragg, Meredith Ringel Morris, Maarten W Bos, Isabelle Albi, and Andrés Monroy-Hernández. 2020. Social app accessibility for deaf signers. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW2 (2020), 1–31.
- [44] Kelly Mack, Emma McDonnell, Venkatesh Potluri, Maggie Xu, Jailyn Zabala, Jeffrey Bigham, Jennifer Mankoff, and Cynthia Bennett. 2022. Anticipate and adjust: Cultivating access in human-centered methods. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–18.
- [45] Kelly Avery Mack, Rida Qadri, Remi Denton, Shaun K Kane, and Cynthia L Bennett. 2024. “They only care to show us the wheelchair”: disability representation in text-to-image AI models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–23.
- [46] H Masoudi. 2024. Effectiveness of ChatGPT in improving English writing proficiency among Non-native English speakers. *International Journal of Educational Sciences and Arts* 3, 4 (2024), 62–84.
- [47] Emma J McDonnell, Ping Liu, Steven M Goodman, Raja Kushalnagar, Jon E Froehlich, and Leah Findlater. 2021. Social, environmental, and technical: Factors at play in the current use and future design of small-group captioning. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW2 (2021), 1–25.
- [48] Emma J McDonnell, Soo Hyun Moon, Lucy Jiang, Steven M Goodman, Raja Kushalnagar, Jon E Froehlich, and Leah Findlater. 2023. “Easier or Harder, Depending on Who the Hearing Person Is”: Codesigning Videoconferencing Tools for Small Groups with Mixed Hearing Status. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–15.
- [49] Brian T McMahon. 2012. An overview of workplace discrimination and disability. *Journal of Vocational Rehabilitation* 36, 3 (2012), 135–139.
- [50] Cecily Morrison, Martin Grayson, Rita Faia Marques, Daniela Massiceti, Camilla Longden, Linda Wen, and Edward Cutrell. 2023. Understanding Personalized Accessibility through Teachable AI: Designing and Evaluating Find My Things for People who are Blind or Low Vision. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility*. 1–12.
- [51] Sarah Morrison and Rhonda Voight-Campbell. 2017. What is ProTactile and what are its benefits? <https://www.tsbvi.edu/tx-senseabilities/issues/fall-2017/what-is-protactile-and-what-are-its-benefits#:~:text=ProTactile%20is%20a%20language%20that,aj%20granda%20and%20jelica%20Nuccio>
- [52] Lucie Naert, Caroline Larboulette, and Sylvie Gibet. 2020. A survey on the animation of signing avatars: From sign representation to utterance synthesis. *Computers & Graphics* 92 (2020), 76–98.
- [53] OpenAI. 2024. How to Change Your Language Setting in ChatGPT. <https://help.openai.com/en/articles/8357869-how-to-change-your-language-setting-in-chatgpt> Accessed: 2024-02-28.
- [54] Noel O’Connell. 2022. “Opportunity blocked”: Deaf people, employment and the sociology of audism. *Humanity & society* 46, 2 (2022), 336–358.
- [55] Austin Pack and Jeffrey Maloney. 2023. Potential affordances of generative ai in language education: Demonstrations and an evaluative framework. *Teaching English with Technology* 23, 2 (2023), 4–24.
- [56] Ilias Papastratis, Christos Chatzikonstantinou, Dimitrios Konstantinidis, Kosmas Dimitropoulos, and Petros Daras. 2021. Artificial intelligence technologies for sign language. *Sensors* 21, 17 (2021), 5843.
- [57] Ila Parasnis. 1998. *Cultural and language diversity and the deaf experience*. Cambridge University Press.
- [58] Sen Qi and Ross E Mitchell. 2012. Large-scale academic achievement testing of deaf and hard-of-hearing students: Past, present, and future. *Journal of deaf studies and deaf education* 17, 1 (2012), 1–18.
- [59] Susan Rose, Patricia L McAnally, and Stephen P Quigley. 2004. *Language learning practices with deaf children*. ERIC.
- [60] Georgia Robins Sadler, Hau-Chen Lee, Rod Seung-Hwan Lim, and Judith Fullerton. 2010. Recruitment of hard-to-reach population subgroups via adaptations of the snowball sampling strategy. *Nursing & health sciences* 12, 3 (2010), 369–374.
- [61] Sven Schild and Constance J Dalenberg. 2012. Trauma exposure and traumatic symptoms in deaf adults. *Psychological trauma: Theory, research, practice, and policy* 4, 1 (2012), 117.
- [62] Sven Schild and Constance J Dalenberg. 2016. Information deprivation trauma: Definition, assessment, and interventions. *Journal of aggression, maltreatment & trauma* 25, 8 (2016), 873–889.
- [63] Jessica A Scott, Hannah M Dostal, and Susan Lane-Outlaw. 2021. A Call for a Diversity of Perspectives in Deaf Education Research. *American Annals of the Deaf* 166, 1 (2021), 49–61.
- [64] Per Koren Solvang and Hilde Haualand. 2014. Accessibility and diversity: Deaf space in action. *Scandinavian Journal of Disability Research* 16, 1 (2014), 1–13.
- [65] Thad Starner, Sean Forbes, Matthew So, David Martin, Rohit Sridhar, Gururaj Deshpande, Sam Sepah, Sahir Shahryar, Khushi Bhardwaj, Tyler Kwok, et al. 2024. PopSign ASL v1.0: an isolated american sign language dataset collected via smartphones. *Advances in Neural Information Processing Systems* 36 (2024).
- [66] William C Stokoe Jr. 2005. Sign language structure: An outline of the visual communication systems of the American deaf. *Journal of deaf studies and deaf education* 10, 1 (2005), 3–37.
- [67] Samuel J Supalla, Jody H Cripps, and Andrew PJ Byrne. 2017. Why American sign language gloss must matter. *American annals of the deaf* 161, 5 (2017), 540–551.
- [68] M Virginia Swisher. 1989. The language-learning situation of deaf students. *Tesol Quarterly* 23, 2 (1989), 239–257.

- [69] Lida Theodorou, Daniela Massiceti, Luisa Zintgraf, Simone Stumpf, Cecily Morrison, Edward Cutrell, Matthew Tobias Harris, and Katja Hofmann. 2021. Disability-first dataset creation: lessons from constructing a dataset for teachable object recognition with blind and low vision data collectors. In *Proceedings of the 23rd International ACM SIGACCESS Conference on Computers and Accessibility*. 1–12.
- [70] Clayton Valli. 2000. *Linguistics of American Sign Language: An Introduction*. Gallaudet University Press.
- [71] Kimberly A Wolbers, Shannon C Graham, Hannah M Dostal, and Lisa M Bowers. 2014. A description of ASL features in writing. *Ampersand* 1 (2014), 19–27.

A Text AI Tools' Responses to the Prompt "Dumb meaning?"

See Figure 2 and Figure 3.

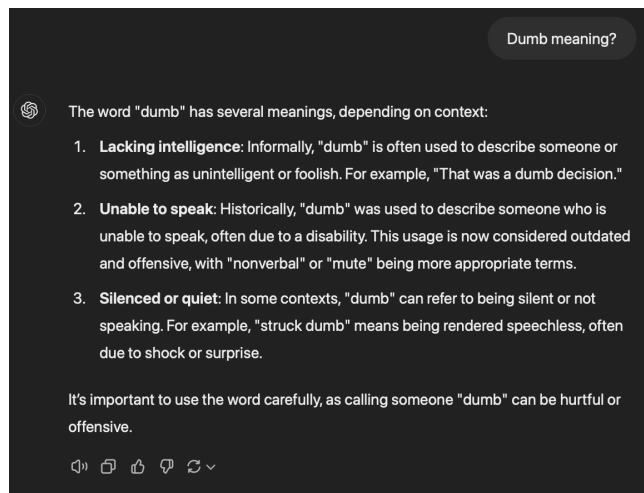


Figure 2: ChatGPT's Response to the Prompt "Dumb meaning?"

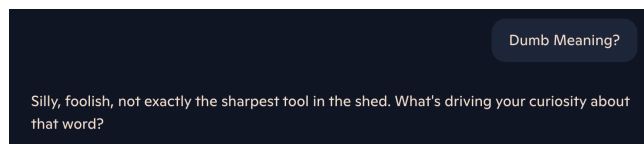


Figure 3: Copilot's Response to the Prompt "Dumb meaning?"

B Survey Questions

Definition: **Text AI**, short for text-based generative AI, is a type of computer program that can read and write human languages. Users interact with Text AI by inputting text prompts, to which the AI responds with generated text. Common examples are OpenAI's ChatGPT, Microsoft's Copilot, and Google's Gemini.

Section 1: Usage of Text AI

1. How often do you use Text AI (please see Text AI's definition above)?

- (1) Daily
- (2) Several times a week
- (3) Rarely
- (4) Never (redirect to **Section 6**, and then **Section 5**)

2. Which Text AI do you usually use? (Select all that apply)

- (1) OpenAI ChatGPT
- (2) Microsoft Copilot
- (3) Google Gemini
- (4) Meta Llama
- (5) Other (please specify)

3. For what purposes do you use Text AI? You can use either ASL or English to answer the question. (Open ended)

4. If you use Text AI for English reading or writing related tasks, what are they? (Select all that apply)

- (1) Writing emails
- (2) Text messaging
- (3) Social media posts
- (4) Work-related communication
- (5) School-related communication
- (6) Other (please specify)
- (7) I do not use Text AI for English-related tasks

Section 2: English Proficiency Improvement

1. What is your comfort level using written English to communicate **without** using Text AI?

- (1) Very uncomfortable
- (2) Somewhat uncomfortable
- (3) Fair
- (4) Comfortable
- (5) Very comfortable

2. What is your comfort level using written English to communicate **with** Text AI?

- (1) Very uncomfortable
- (2) Somewhat uncomfortable
- (3) Fair
- (4) Comfortable
- (5) Very comfortable

3. When you use Text AI, do you specify your English level in your initial chat with the Text AI (aka prompt)? For example, "English is not my first language so please simplify your answer". You can use either ASL or English to answer the question. (Open ended)

4. Other than the "trick" above, do you have other "tricks" that you use to make the Text AI generate something more suitable for you? You can use either ASL or English to answer the question. (Open ended)

5. Which aspects of your English writing have improved (if any) while using Text AI? (Select all that apply)

- (1) Topics and ideas
- (2) Organization and logical order
- (3) Tone and voice (conveying the writer's feelings, attitude, and emotion)
- (4) Vocabulary and word choice
- (5) Sentence fluency and readability
- (6) Conventions (rule of spelling, punctuation, capitalization, and grammar)
- (7) Overall confidence in writing
- (8) Other (please specify)

Section 3: Communication Barriers and Discrimination

1. **Without** using Text AI, how often do you feel anxious / nervous when communicating with hearing people in writing?

- (1) Daily
- (2) Several times a week
- (3) Rarely
- (4) Never

2. **With** Text AI, how often do you feel anxious / nervous when communicating with hearing individuals in writing?

- (1) Daily
- (2) Several times a week
- (3) Rarely
- (4) Never

3. Do you feel that you have experienced increased or reduced discrimination since using Text AI?

- (1) Significantly decreased (much less discrimination)
- (2) Somewhat decreased (less discrimination)
- (3) No change
- (4) It has increased (more discrimination)
- (5) I can't tell

Section 4: Satisfaction and Suggestions

1. How satisfied are you with the assistance provided by Text AI in improving your English proficiency?

- (1) Very satisfied
- (2) Satisfied
- (3) Neutral
- (4) Dissatisfied
- (5) Very dissatisfied

2. Following up on your answer above: why or why not are you very satisfied / satisfied / neutral / dissatisfied / very dissatisfied with the assistance provided by Text AI? You can use either ASL or English. (Open ended)

3. What difficulties have you encountered while using Text AI? (Select all that apply)

- (1) Difficulty understanding responses
- (2) Limited sign language support
- (3) Difficulty in asking questions / finding the right prompts
- (4) Technical issues, such as network connectivity issues
- (5) Not sure how Text AI can help
- (6) Other (please specify)

4. Do you have any suggestions for improving Text AI assistance for Deaf/HoH users? You can either use ASL or English. (Open ended)

Section 5: Demographic Information

1. What is your age?

- (1) 18-25
- (2) 26-35
- (3) 36-50
- (4) 51 and above

2. What is your highest level of education?

- (1) High school / GED or less
- (2) Some college
- (3) Associate degree or equivalent
- (4) Bachelor's degree
- (5) Graduate degree

(6) Other (please specify)

3. What is your identity in the DHH community?

- (1) Deaf
- (2) deaf
- (3) Hard of hearing
- (4) DeafBlind
- (5) Other (please specify)

4. What is your first language? (Select all that apply)

- (1) English
- (2) ASL
- (3) Other (please specify)

5. What is your primary language for communication?

- (1) English
- (2) ASL
- (3) Other (please specify)

6. We are thinking of running interview sessions with a subset of this survey's respondents. Would you like to be contacted about this opportunity?

- (1) Yes
- (2) No

7. What's your email address?

Section 6: Possibility of using Text AI (for people who have never used it)

1. What are the main reasons you have not used Text AI? (Select all that apply)

- (1) Not aware of text AI tools
- (2) Privacy concerns
- (3) Lack of trust in AI
- (4) Difficulty in using technology
- (5) Prefer human assistance
- (6) Not sure what Text AI can do
- (7) Other (please specify)

2. Would you consider using Text AI in the future?

- (1) Yes
- (2) No
- (3) Unsure

3. If you were to consider using Text AI, what tasks do you think you might use them for? (Select all that apply, and please think beyond AI's current abilities)

- (1) Improving English writing skills
- (2) Translating ASL to English
- (3) Translating English to ASL
- (4) Learning new vocabulary
- (5) Practicing English grammar
- (6) Reducing anxiety before communicating with hearing people
- (7) Other (please specify)

4. What features or improvements would make you more likely to use Text AI? You can either use ASL or English. (Open ended)

Received 23 January 2025; accepted 20 February 2025